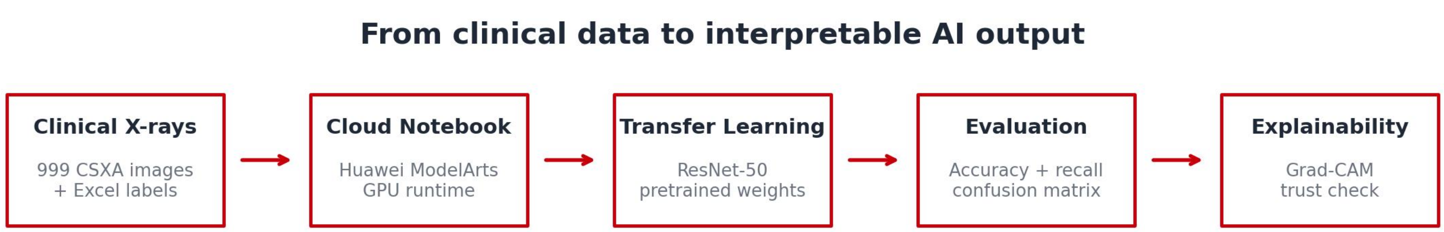
Medical Image Classification with ResNet-50

Laboratory Note

Instructor: Dong Chenxi

Prepared for Year 1-2 undergraduate students new to practical deep learning

With special thanks to Prof. Philip Yu for his valuable advice on the pedagogical design.



Learning Objectives

1. Operate a GPU-powered notebook on Cloud.
2. Construct a PyTorch data pipeline with stratified splitting and medically-appropriate augmentation.
3. Apply transfer learning with a pretrained ResNet-50 and implement best-practice training (early stopping, LR scheduling).
4. Evaluate the model using accuracy, per-class precision/recall/F1, and a confusion matrix rather than relying on one headline metric.
5. Use Grad-CAM to check whether the model focuses on the cervical vertebrae rather than irrelevant image borders or artifacts.
6. Explain the practical limitations of a small medical dataset and propose one evidence-based improvement after the lab.

1. Why This Lab Matters

Many students and IT professionals spend long hours in front of digital screens, and neck discomfort is a common clinical complaint in this population. When cervical spine alignment needs to be assessed, X-ray imaging is one widely used clinical tool. In this lab, we use cervical spine X-rays to study how deep learning can support curvature classification.

⚠ *Important: this lab is for education and research training only. The model we build is not a clinical diagnostic system and should not be used for patient care.*

We utilize the **Cervical Spine X-ray Atlas (CSXA)** dataset — 999 real, expert-annotated clinical images. The task is framed as a 4-class problem: **Lordotic (normal), Straight, Sigmoid, and Kyphotic**.

> **Key Concept: Authentic Learning vs. Toy Datasets**
>
> Most academic tutorials rely on sanitized datasets (MNIST). These fail to capture real-world clinical challenges: severe class imbalance, subtle inter-class visual boundaries, and label ambiguity between Straight and Sigmoid curvatures. The CSXA dataset exposes students to these authentic difficulties from day one.

Modern machine learning is rarely performed on local laptops. This lab standardizes the environment using Huawei Cloud, providing every student with an identical NVIDIA Tesla V100 32 GB GPU instance. This mirrors real industry workflows and eliminates "works on my machine" friction.

2. Preparation: Notebook Setup

Huawei Cloud ModelArts is used because it gives the class a standardized GPU notebook environment. This removes a common teaching problem: different laptops, different drivers, different Python versions, and different results.

1. Open the notebook instance.
2. Select a PyTorch environment with GPU support.
3. Open JupyterLab and locate the provided Kaggle/ModelArts notebook.
4. Run the environment-check cell before loading data.

```
import torch
print('torch cuda available:', torch.cuda.is_available())
print('torch version:', torch.__version__)
!nvidia-smi
```

3. Dataset: Loading Images and Matching Labels

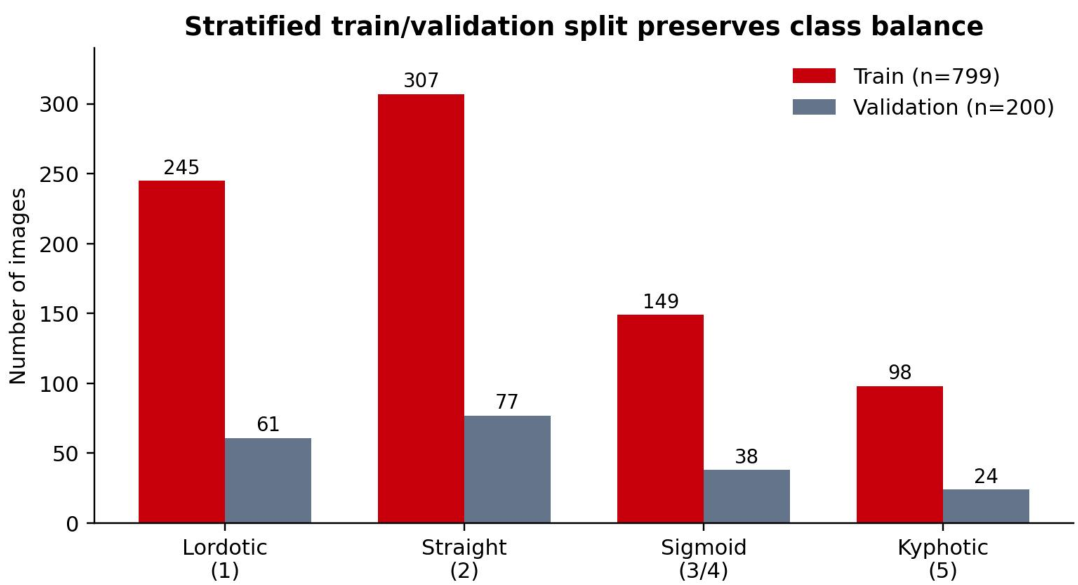
The notebook downloads the image folder and the Excel label file. Each image filename begins with a four-digit identifier. The Excel sheet contains the corresponding image number and curvature label. The first practical task is therefore not modelling - it is reliable data joining.

```
IMAGE_DIR = kagglehub.dataset_download('ddatad/cervical-x-ray')
LABEL_ROOT = kagglehub.dataset_download('ddatad/dataset')
LABEL_FILE = os.path.join(LABEL_ROOT, 'dataset.xlsx')

df = pd.read_excel(LABEL_FILE, engine='openpyxl')
df['Curvature'] = df['Curvature'].replace({4: 3})  # merge Sigmoid1 and Sigmoid2
```

The lab merges Sigmoid1 and Sigmoid2 into one Sigmoid class. This is a teaching design choice: the two subtypes are visually subtle.

We perform a stratified 80/20 train-validation split to preserve class proportion

Figure 1: Stratified Train/Validation Split (Total 999 Images)



Straight is the majority class; Kyphotic is the minority. Stratification ensures both splits reflect this distribution.

4. Image Transforms: Gentle Augmentation, Not Random Distortion

Augmentation is used to make the model less sensitive to small changes in patient positioning. However, medical augmentation must be conservative. We want to simulate plausible acquisition differences, not create anatomically unrealistic images.

Transform	Why we use it	Practical caution
Resize + crop to 224x224	ResNet-50 expects a standard input size.	Avoid cropping away the cervical spine.
Random rotation ±10°	Simulates minor positioning differences.	Large rotations can create unrealistic anatomy.
Horizontal flip	Adds diversity in a simple way in this teaching task.	In some medical tasks left/right orientation matters; check with clinicians.
ImageNet normalization	Matches the scale expected by pretrained ResNet-50.	Do not remove it unless you retrain/retune carefully.

```
train_tfms = T.Compose([
    T.Resize((256, 256)),
    T.RandomResizedCrop(224, scale=(0.85, 1.0)),
    T.RandomHorizontalFlip(),
    T.RandomRotation(10),
    T.ToTensor(),
    T.Normalize([0.485,0.456,0.406], [0.229,0.224,0.225])
])
```

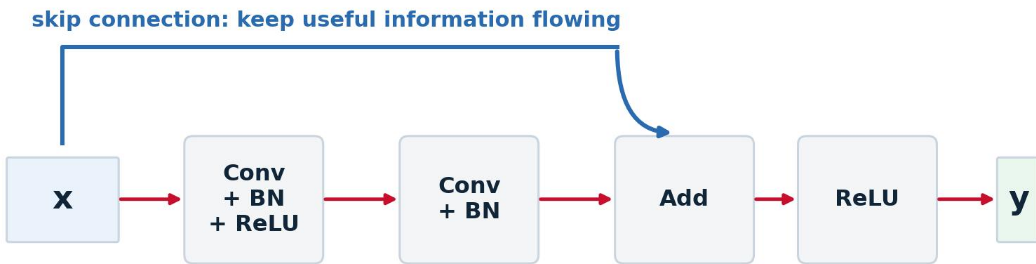
5. Model: Fine-Tuning ResNet-50

Training a deep network from scratch on 999 images would usually overfit. Transfer learning is the practical solution: start from a model that already learned general visual features from a large image dataset, then adapt it gently to the cervical X-ray task.

The Power of Skip Connections

The defining innovation of ResNet is the residual (skip) connection. The key idea of ResNet is simple: instead of forcing every layer to completely transform the input, ResNet gives information a shortcut path. If a few layers do not improve the representation, the network can still pass useful information forward through the skip connection. In practice, this makes deep networks easier to train and makes fine-tuning more stable.

Figure 2: In ResNet-50 Skip connections help information flow through deep networks



Why this matters in lab: ResNet-50 is deep, but skip connections make fine-tuning more stable on a small medical image dataset.

Why we do it this way
The pretrained model already knows useful low-level visual features such as edges, contrast changes, and texture patterns. Fine-tuning adapts those features to the medical image task with fewer examples.

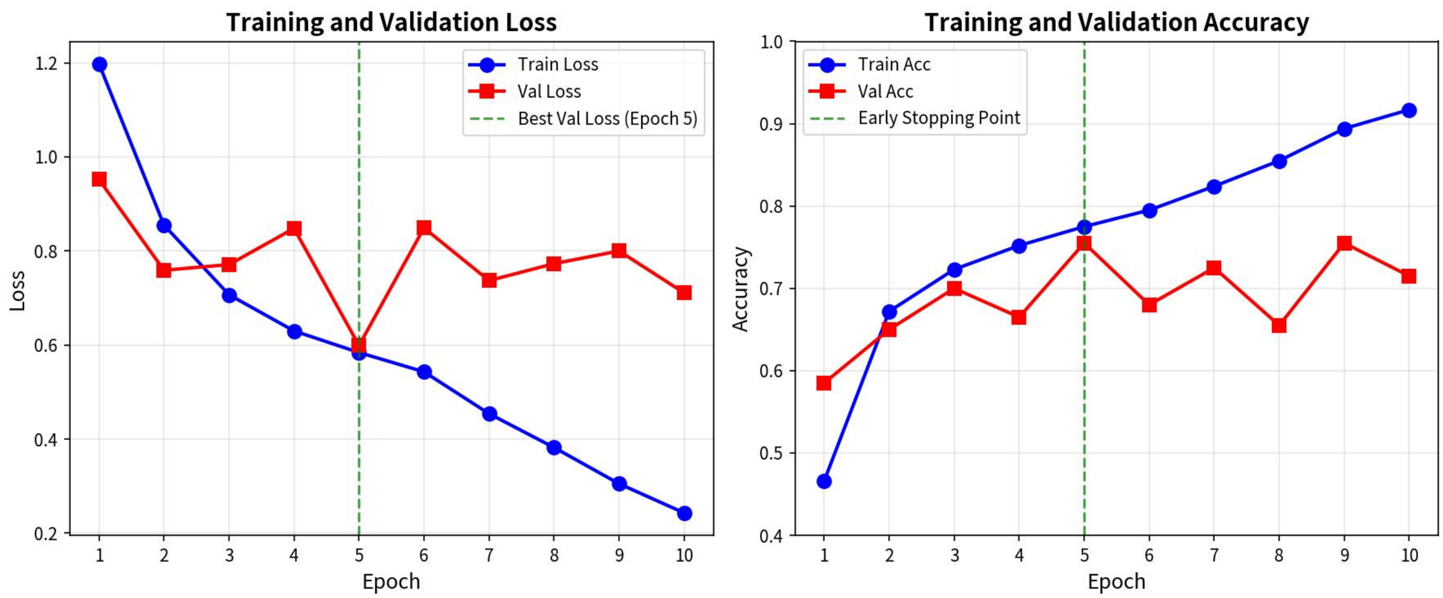
```
model = models.resnet50(weights=models.ResNet50_Weights.IMAGENET1K_V1)
model.fc = nn.Linear(model.fc.in_features, num_classes)
model = model.to(device)

criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=1e-4, weight_decay=1e-4)
scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(
    optimizer, mode='min', factor=0.5, patience=2)
```

We use a deliberately small learning rate (1e-4) and weight decay (1e-4). This protects the valuable low-level features (edges, bone textures) already learned by the ImageNet-pretrained backbone while allowing gentle adaptation to X-ray data

6. Training: What to Watch During the Run

During training, students should not only wait for the final score. They should monitor whether training loss and validation loss move together. When training improves but validation gets worse, the model is starting to memorize the training set.

Figure 3: Training and Validation Curves (Typical Run)



Validation loss reaches its minimum at Epoch 5. Continued training causes overfitting — early stopping automatically restores the best checkpoint.

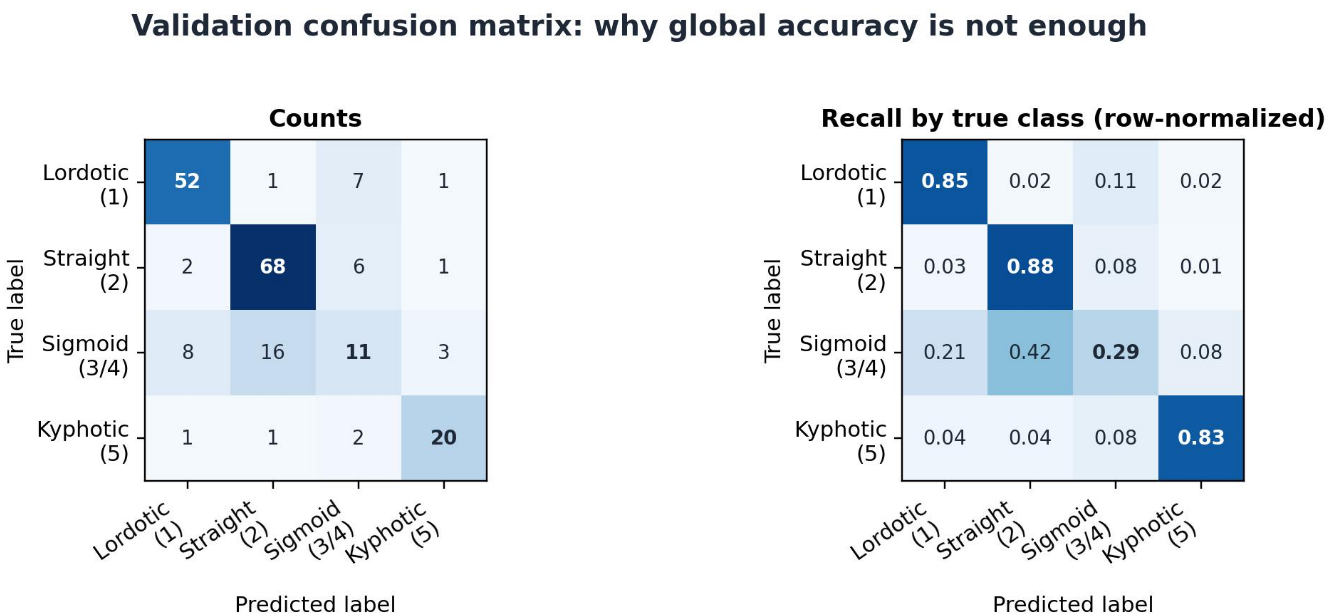
7. Holistic Evaluation: Confusion Matrix and Class-Level Metrics

In clinical AI, global accuracy is a dangerously incomplete metric. We must examine performance at the per-class level.

Key Concept: The Sigmoid Failure Case

While Straight achieves 88% recall and Kyphotic 83%, the Sigmoid class collapses to only 29% recall. This is caused by two factors: (1) genuine visual ambiguity — the physiological boundary between Straight and Sigmoid is subtle even to expert radiologists, and (2) data imbalance — the model is biased toward majority classes. Global accuracy masks this critical weakness.

Figure 4: Validation confusion matrix for the baseline ResNet-50 model. Left: raw counts. Right: row-normalized values, which can be read as recall for each true class.



Row-normalized values = recall per class. Notice the severe under-performance on Sigmoid (only 29% of true Sigmoid cases correctly identified).

The model achieves 75.5% validation accuracy, but the row-normalized matrix shows the real issue: Lordotic, Straight, and Kyphotic have high recall, while Sigmoid is much harder.

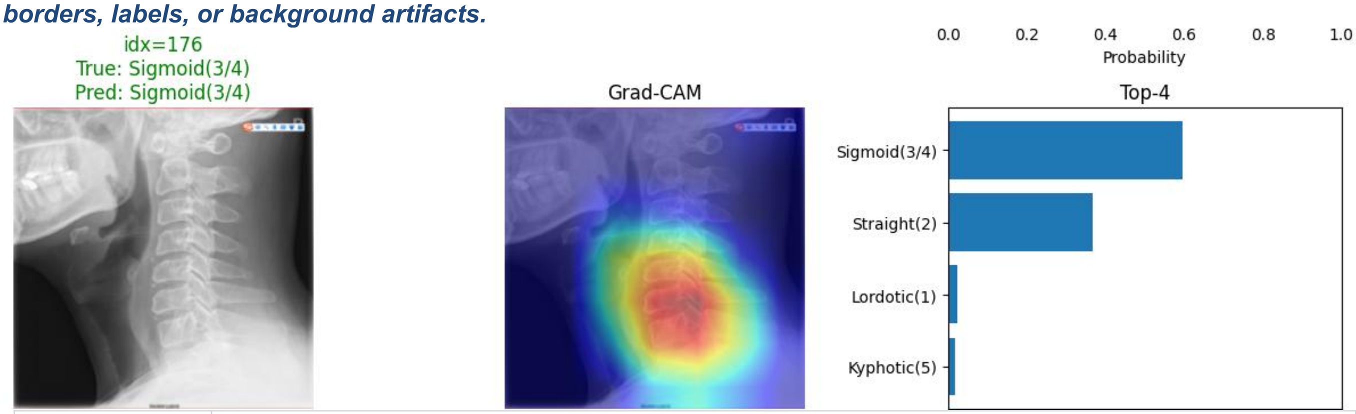
8. Model Interpretability: Opening the Black Box with Grad-CAM

Deploying black-box models in healthcare carries ethical and safety risks. We must verify that the network attends to the same anatomical structures a radiologist would examine. We therefore apply **Gradient-weighted Class Activation Mapping (Grad-CAM)** to the final convolutional block.

```
heatmap = grad_cam(
    model,
    img.unsqueeze(0).to(device),
    class_idx=pred,
    target_layer='layer4')
```

Grad-CAM computes the gradient of the target class score with respect to the feature maps, then produces a spatial heatmap overlaid on the original X-ray. A trustworthy model exhibits strong activation (red/yellow zones) localized precisely over the cervical vertebrae, ignoring irrelevant background or artifacts.

Figure 5. Grad-CAM is used as a sanity check: the activation should fall over the cervical vertebrae, not over image borders, labels, or background artifacts.



Good sign	Activation is concentrated on the cervical vertebrae and curvature-relevant anatomy.
Warning sign	Activation is mostly on image borders, text markers, background, or unrelated artifacts.

⚠ The Grad-Cam Heatmap is a sanity check, not a proof of clinical reasoning. It shows where the model is looking, but it cannot tell you what the model is concluding. A model might look at the correct cervical vertebra but still make the wrong curvature classification due to poor feature extraction. Treat Grad-CAM as one piece of debugging evidence, not the final answer.

9. In-Class Deliverables

Submit the following before leaving the lab

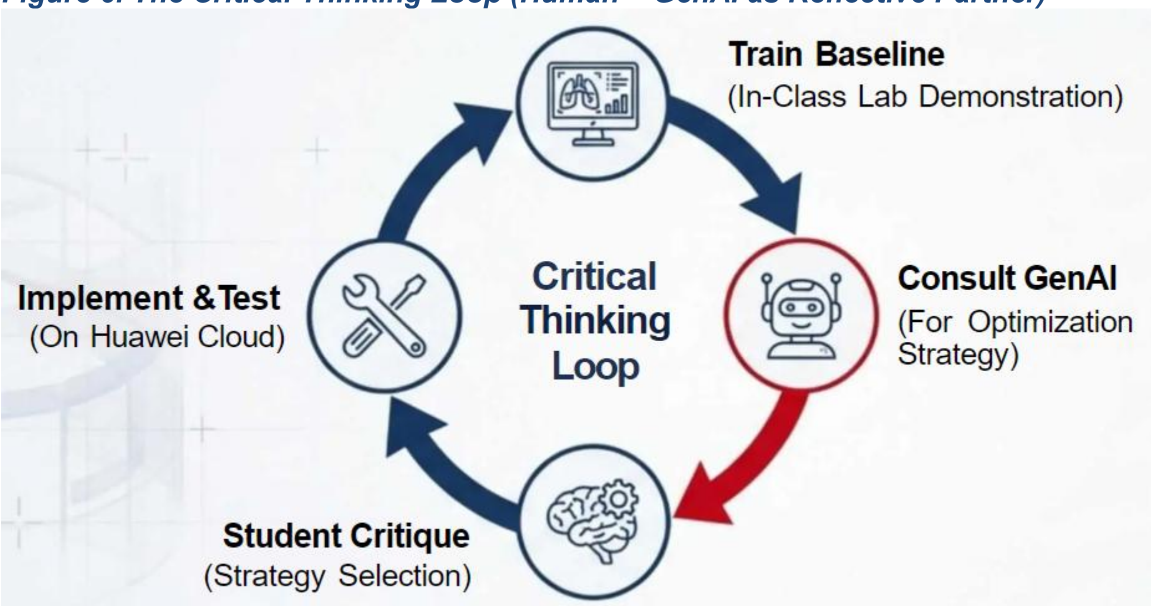
Done	Task
[ ]	Screenshot of train and validation class distributions.
[ ]	Screenshot learning curve figure or printed epoch logs.
[ ]	Screenshot the Confusion matrix and classification report.
[ ]	Pick one Grad-CAM example with your interpretation.
[ ]	Identify the weakest class and explain one possible reason.

10. After-Class Inquiry Task

The baseline model struggles with the Sigmoid class. Your after-class task is to act as the lead engineer and propose one evidence-based improvement. You may consult a Generative AI assistant such as ChatGPT or DeepSeek for possible strategies, such as class-weighted loss, oversampling, stronger augmentation, or threshold calibration. However, you must critique the suggestion before using it. Select one strategy, implement it, retrain the model, and compare the result with the baseline.

Tip: GenAI as a Partner, Not a Shortcut!
Do not just copy - paste code. The goal is human - in - the - loop engineering. Use GenAI to brainstorm, but use your critical thinking to decide which strategy actually makes sense for medical imaging.

Figure 6. The Critical Thinking Loop (Human + GenAI as Reflective Partner)



Want to Learn More?

- *ResNet & Skip Connections:*
  - *Original Paper: https://arxiv.org/pdf/1512.03385.pdf*
  - *Visual Explanation: https://d2l.ai/chapter_convolutional-modern/resnet.html*
- *Transfer Learning: https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html)*
- *Grad-CAM: https://arxiv.org/pdf/1610.02391.pdf*

Full code: github.com/098765d/Huawei-ICT-Teaching-Competition-2025-26
Dataset: Ran et al. (2024). A high-quality dataset featuring classified and annotated cervical spine X-ray atlas. Scientific Data